

FILTER DESIGN TEST CASE

Duplexer Module - WiFi 2.4 GHz for 4G LTE Carrier Aggregation
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Author:	Jennifer Kim
Reviewer:	Mark Sullivan
Design Center:	Osaka Design Center
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Duplexer Module filter targeting WiFi 2.4 GHz for 4G LTE Carrier Aggregation. The filter is designed on AIN on SiC substrate with a target center frequency of 5194.9 MHz and bandwidth of 64.7 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 4G LTE Carrier Aggregation module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Duplexer Module
- Frequency Band: WiFi 2.4 GHz
- Application: 4G LTE Carrier Aggregation
- Substrate: AIN on SiC
- Package: QFN (Quad Flat No-Lead)
- Design Tool: Ansys HFSS 2024R1
- Design Center: Osaka Design Center

This specification applies to all variants of the Duplexer Module design targeting WiFi 2.4 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5194.9 MHz
Bandwidth (3 dB):	64.7 MHz
Insertion Loss Target:	<= 2.1 dB
Return Loss Target:	>= 13.0 dB
Out-of-Band Rejection:	>= 33.0 dB
Isolation (Duplexer):	>= 36.0 dB
Q Factor:	2952
Temperature Coefficient:	-27.4 ppm/°C
Die Size:	2.6 x 0.6 mm
Wafer Size:	8-inch
Number of Resonators:	6
Electrode Material:	Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Ansys HFSS 2024R1 and load the Duplexer Module template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 5194.9 MHz, BW 64.7 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on SiC).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15585 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.1 dB
- Return loss in passband shall be >= 13.0 dB
- Out-of-band rejection at +/- 97 MHz offset >= 33.0 dB
- Group delay variation across passband shall be <= 7 ns
- Temperature drift over -40°C to +85°C shall be <= 17.8 MHz
- Power handling shall be >= +27 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2025-04-07
Devices Tested: 94
Insertion Loss (meas): 1.8 dB
Return Loss (meas): 15.0 dB
Rejection (meas): 36.1 dB
Isolation (meas): 40.8 dB
Q Factor (meas): 2952
Group Delay Variation: 6.4 ns
Power Handling: +29 dBm
Die Yield: 94.7%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Duplexer Module design demonstrated below-target performance for WiFi 2.4 GHz.
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) ESD robustness exceeded specification by 2x margin.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

Author:	Jennifer Kim	Date:	2025-04-07
Reviewer:	Mark Sullivan	Date:	2025-04-09
Approver:	Dr. James Liu	Date:	2025-04-10

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